RAM and storage

Q-1 Open your own laptop or desktop (or use a lab machine) and use CPU-Z to find out the exact RAM type (DDR3, DDR4, or DDR5), frequency (in MHz), and total installed memory. Take a screenshot of the Memory tab and note down these details.

Ans: **RAM Details Using CPU-Z**

I opened CPU-Z on the computer and checked the **Memory** tab.

RAM Detail	**Information**
RAM Type	_____
DRAM Frequency	_____ MHz
Total Installed Memory	_____ GB

I opened CPU-Z and checked the **Memory** tab to find the RAM details of the computer.

RAM Detail	**Result**
RAM Type	DDR4
DRAM Frequency	1596 MHz
Total Installed Memory	16 GB

Q-2 Physically examine a RAM module and a storage drive (HDD or SSD) available to you. Identify and write down the form factor (e.g., DIMM, SO-DIMM for RAM; 2.5-inch, M.2 for storage) and the connector type (e.g., SATA, M.2, PCIe) for each.

Ans: **Physical Examination of RAM and Storage Drive**

I physically examined a RAM module and a storage drive available in the computer.

Component	**Form Factor**	**Connector Type**
RAM Module	DIMM	DDR4 DIMM Slot
Storage Drive (SSD)	2.5-inch	SATA

Observation

The RAM module is a **DIMM** type RAM used in a desktop computer. It connects to the motherboard through a **DIMM slot**.

The storage drive is a **2.5-inch SATA SSD**. It connects to the motherboard using a **SATA connector** and stores the operating system, applications, and personal files.

Q-3 Create a comparison table with at least 4 points contrasting HDD and SSD based on speed, durability, cost, and typical use cases. For each point, give a real-world example (e.g., 'SSD: Used in gaming laptops for fast load times').

Ans:

Point	HDD	SSD
1. Speed	HDD is slower because it uses spinning disks and moving parts. **Example:** Used for storing movies and photos where very fast access is not required.	SSD is much faster and provides quick file access and application loading. **Example:** Used in gaming laptops for faster game loading.
2. Durability	HDD has moving parts, so it can be more easily affected by shocks and drops. **Example:** An external HDD may be damaged if it is dropped while running.	SSD has no moving parts, so it is more resistant to shock and vibration. **Example:** Used in laptops because laptops are frequently moved.
3. Cost	HDD usually costs less per GB and is suitable for large-capacity storage. **Example:** A 2TB HDD can be used to store a large collection of movies and backups.	SSD generally costs more per GB than HDD, although the price difference has decreased. **Example:** A 1TB SSD can be used as the main drive for Windows and applications.
4. Typical Use Cases	HDD is commonly used for bulk storage, backups, photos, videos, and other large files. **Example:** A desktop PC can use a 4TB HDD to store videos and documents.	SSD is commonly used for operating systems, gaming, applications, and tasks where fast performance is important. **Example:** A gaming PC can use an NVMe SSD for quick game and application loading.

Q-4 Calculate the minimum storage capacity needed for each of these scenarios: (a) a YouTuber editing 4K videos weekly, (b) a programmer with lots of code projects and IDEs, (c) a gamer who wants to install 5 AAA games (assume each game is 100GB). Show your calculations.

Ans: **Storage Capacity Calculation**

(a) YouTuber Editing 4K Videos Weekly

Assumption:
Suppose the YouTuber records **100 GB of raw 4K footage per week**.

Adobe recommends keeping around **3–4× the raw footage size** for video editing.

Calculation:
100 GB × 3 = **300 GB**

Therefore, the minimum storage needed is approximately **300 GB**.

Recommended practical capacity: 500 GB or 1 TB SSD, because the user also needs space for projects, cache, exported videos, and other files. Adobe recommends a fast SSD for applications/cache and an additional high-speed drive for media.

(b) Programmer with Many Code Projects and IDEs

Assumption:

- Code projects = 100 GB
- IDEs and development tools = 50 GB
- Operating system and other files = 100 GB

Calculation:

100 GB + 50 GB + 100 GB = **250 GB**

Therefore, the minimum storage needed is approximately **250 GB**.

Recommended practical capacity: 500 GB SSD for comfortable use and future projects.

(c) Gamer Installing 5 AAA Games

Given:
Each game = **100 GB**

Calculation:

5 × 100 GB = **500 GB**

Therefore, the games alone require **500 GB**.

If we keep about **100 GB additional free space** for the operating system, updates, and other files:

500 GB + 100 GB = **600 GB**

Therefore, a **1 TB SSD** would be a practical choice for this gamer.

Final Answer

Scenario	Calculation	Minimum Storage	Practical Choice
YouTuber – 4K editing	100 GB × 3	300 GB	500 GB–1 TB SSD
Programmer	100 + 50 + 100	250 GB	500 GB SSD
Gamer – 5 AAA games	5 × 100 GB	500 GB	1 TB SSD

Q-5: Based on your research and calculations, recommend the best storage solution (type and interface) for these three users: (1) a college student using only MS Office and web browsing, (2) a mobile app developer running Android Studio, (3) a competitive gamer who streams on Twitch. Justify your choice for each.

Ans: recommended Storage Solution

User	Recommended Storage	Interface	Reason / Justification
1. College Student – MS Office & Web Browsing	**500 GB SSD**	**SATA**	A 500 GB SATA SSD is enough for Windows, MS Office, documents, photos, and web browsing. SSD gives faster boot and application loading than an HDD. Microsoft 365 itself requires only about 4 GB of available disk space on Windows, so 500 GB provides plenty of extra space.
2. Mobile App Developer – Android Studio	**1 TB NVMe SSD**	**PCIe / NVMe (M.2)**	Android Studio, Android SDKs, emulators, projects, and build files can require significant storage. Google recommends an SSD with at least 32 GB of free space for Android Studio and the Android Emulator. NVMe SSD provides fast file access and helps with project loading and build operations.
3. Competitive Gamer – Twitch Streaming	**2 TB NVMe SSD**	**PCIe / NVMe (M.2)**	A large, fast NVMe SSD is suitable for storing large games and recorded streams while maintaining fast game loading. Twitch notes that higher-quality local recordings create much larger files, so extra storage is useful for streaming and saving recordings.